

Lecture 3: Plotting

TECH2: Introduction to Programming, Data, and Information Technology

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October 14, 2026

See GitHub repository for notebooks and data:

<https://github.com/richardfoltyn/TECH2-H26>

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1 Plotting with Matplotlib

In this lecture, we study how to plot numerical data. Python itself does not have any built-in plotting capabilities, so we will be using `matplotlib`, the most popular graphics library for Python.

- For details on a particular plotting function, see the [official documentation](#).
- There is an official introductory [tutorial](#) which you can use alongside the material presented here.

In order to access the functions and objects from Matplotlib, we first need to import them. The general convention is to use the namespace `plt` for this purpose:

```
import matplotlib.pyplot as plt
```

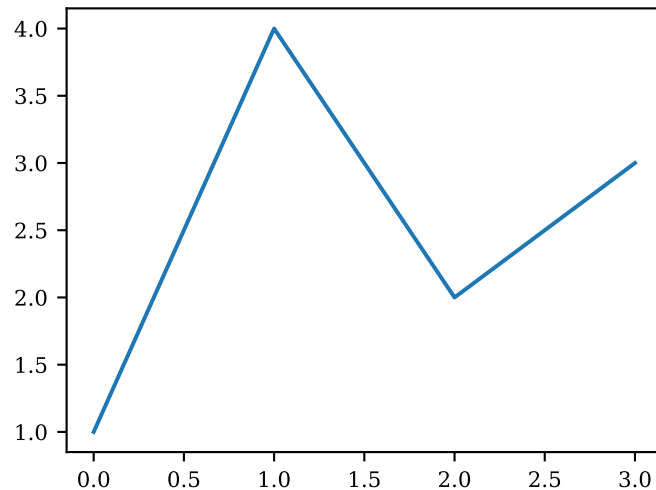
Note that there is an additional high-level plotting library called [seaborn](#) which builds on top of Matplotlib with a focus on providing convenient functions to create statistical graphs.

1.1 Line plots

One of the simplest plots we can generate using the `plot()` function is a line defined by a list of y -values.

```
[1]: # import matplotlib library
import matplotlib.pyplot as plt

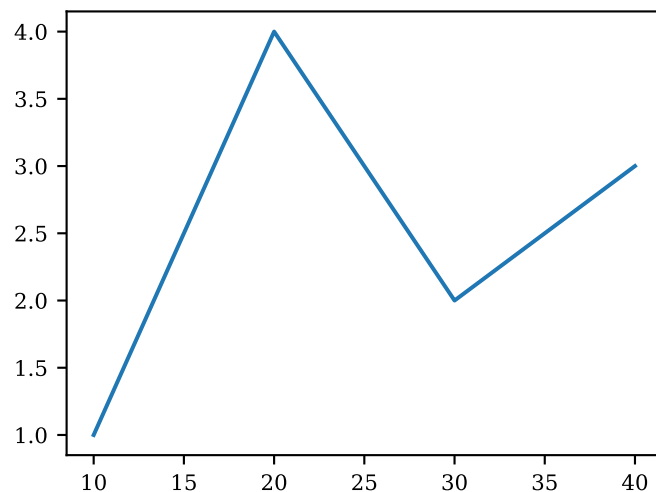
# Plot list of integers
yvalues = [1, 4, 2, 3]
_ = plt.plot(yvalues)
```



Jupyter displays the value of the last expression in a cell. Plotting functions return objects whose text representation is usually unhelpful. Assigning the result to `_` suppresses that extra output while the plot still appears. Here, `_` is simply a conventional name for a result we don't need.

In the plot above, we didn't even have to specify the corresponding x -values, as Matplotlib automatically assumes them to be $[0, 1, 2, \dots]$. Usually, we want to plot for a given set of x -values like this:

```
[2]: # explicitly specify x-values
xvalues = [10, 20, 30, 40]
_ = plt.plot(xvalues, yvalues)
```



Your turn

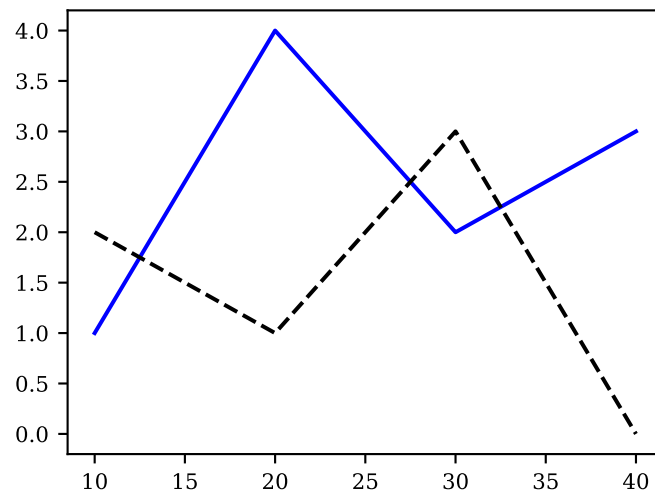
Plot the quadratic function $y = x^2$ on the interval $[-1, 1]$:

1. Create a grid of 50 x -values using `np.linspace()`.
2. Compute the corresponding y -values and plot them.

1.1.1 Plotting multiple lines

We can also specify multiple lines to be plotted in a single graph:

```
[3]: yvalues2 = [2.0, 1.0, 3.0, 0.0]
_ = plt.plot(xvalues, yvalues, 'b-', xvalues, yvalues2, 'k--')
```



1.1.2 Specifying plot styles

The characters following each set of y -values are style specifications. The letters are shorthand notations for colors (see [here](#) for details):

- b: blue
- g: green
- r: red
- c: cyan
- m: magenta
- y: yellow
- k: black
- w: white

The remaining characters set the line styles. Valid values are:

- - solid line
- -- dashed line
- - . dash-dotted line
- : dotted line

Additionally, we can append marker symbols to the style specification. The most frequently used ones are:

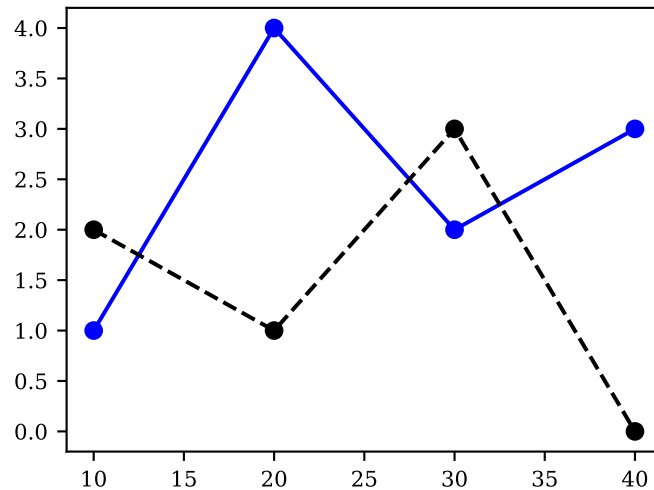
- o: circle
- s: square
- *: star
- x: x

- d: (thin) diamond

The whole list of supported symbols can be found [here](#).

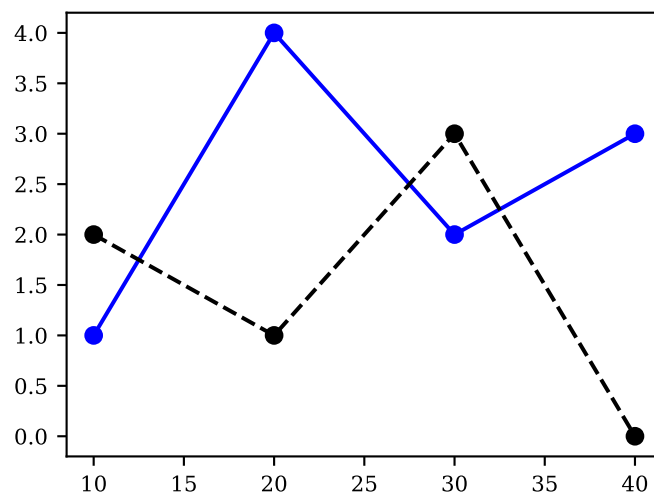
Instead of passing multiple values to be plotted at once, we can also repeatedly call `plot()` to add additional elements to a graph. This is more flexible since we can pass additional arguments which are specific to one particular set of data, such as labels displayed in legends.

```
[4]: # Plot two lines by calling plot() twice
plt.plot(xvalues, yvalues, 'b-o')
_ = plt.plot(xvalues, yvalues2, 'k--o')
```



Individual calls to `plot()` also allow us to specify styles more explicitly using keyword arguments:

```
[5]: # pass plot styles as explicit keyword arguments
plt.plot(xvalues, yvalues, color='blue', linestyle='-', marker='o')
_ = plt.plot(xvalues, yvalues2, color='black', linestyle='--', marker='o')
```



Note that in the example above, we use named colors such as 'blue' or 'black' (see [here](#) for the complete list of named colors).

Matplotlib accepts abbreviations for the most common style definitions using the following shortcuts:

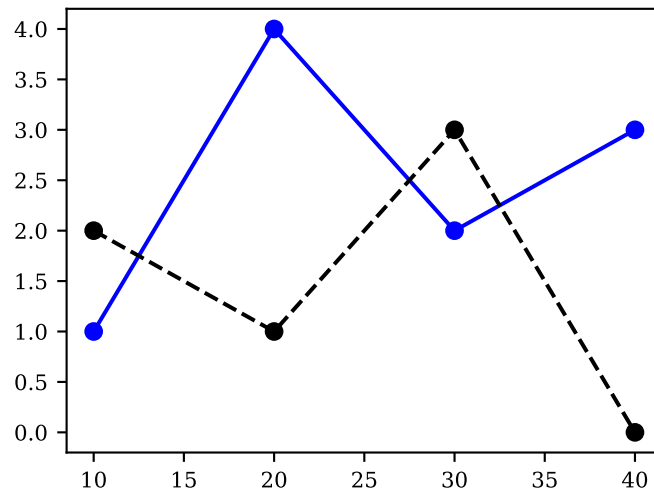
- c or color
- ls or linestyle

- `lw` or `linewidth`
- `ms` or `markersize`

See the section on *Other Parameters* in the `plot()` documentation for a complete list of arguments and their abbreviations.

We can thus rewrite the above code as follows:

```
[6]: # abbreviate plot style keywords
plt.plot(xvalues, yvalues, c='blue', ls='-', marker='o')
_ = plt.plot(xvalues, yvalues2, c='black', ls='--', marker='o')
```



Your turn

Use the data files located in the folder `../data/FRED` to perform the following tasks:

1. Load the data from `FRED_annual.csv`. The file contains annual observations on selected macroeconomic variables for the US.
2. Plot the unemployment rate (column `UNRATE`) using a blue dashed line with line width 0.5 and the inflation rate (column `INFLATION`) using an orange line with line width 0.75 in the *same* figure.

1.2 Scatter plots

We use the `scatter()` function to create scatter plots in a similar fashion to line plots.

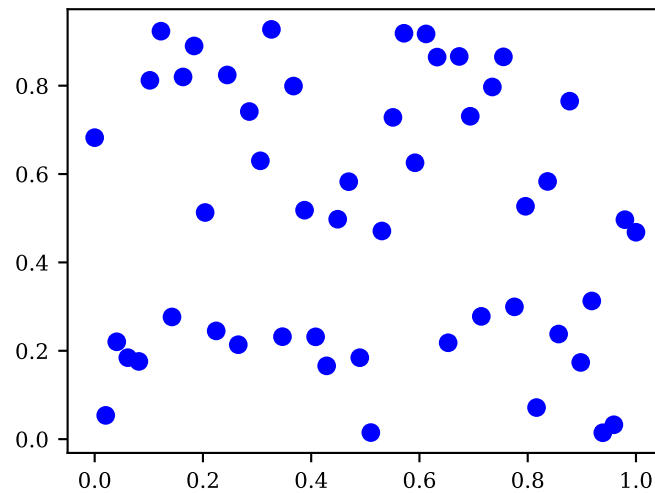
To illustrate, we first generate some random data using the NumPy random number generator (RNG) interface. We won't go into detail about how to generate random numbers with NumPy, so you can just take that code as given.

```
[7]: import matplotlib.pyplot as plt
import numpy as np

# Number of points
N = 50

# Create 50 uniformly spaced values on the unit interval
xvalues = np.linspace(0.0, 1.0, N)
# Draw random numbers from interval [0, 1)
yvalues = np.random.default_rng(seed=123).random(N)
```

```
_ = plt.scatter(xvalues, yvalues, color='blue')
```

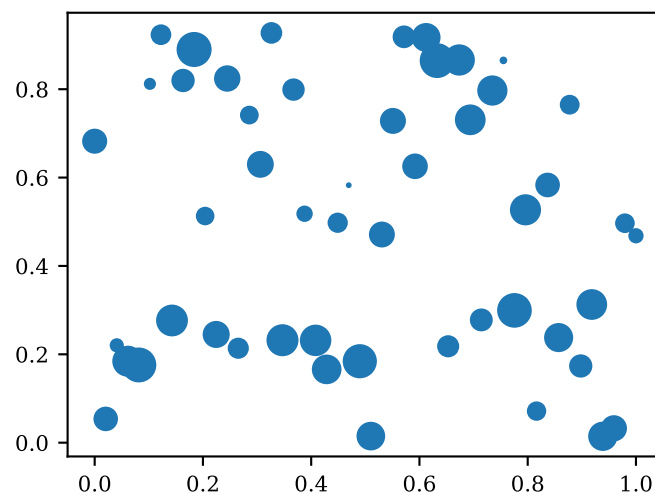


We could in principle create scatter plots using `plot()` by turning off the connecting lines. However, `scatter()` allows us to specify the color and marker size as collections, so we can vary these for every point. `plot()`, on the other hand, imposes the same style on all points within each dataset.

The `s` argument of `scatter()` specifies marker area in points squared. To illustrate, the following code assigns a different (randomly drawn) marker area to each dot:

```
[8]: # Draw random marker areas in points squared
size = np.random.default_rng(seed=456).random(N) * 150.0

# plot with point-specific marker areas
_ = plt.scatter(xvalues, yvalues, s=size)
```



Your turn

Use the data files located in the folder `../..data/FRED` to perform the following tasks:

1. Load the data in `FRED_monthly.csv`. The file contains monthly observations on selected macroeconomic variables for the US.
2. Create a scatter plot of the real interest rate (column `REALRATE`) on the y-axis against the federal funds rate (column `FEDFUNDS`) on the x-axis. Specify the arguments

edgecolors='blue' and color='none' to plot the data as blue rings.

1.3 Histograms

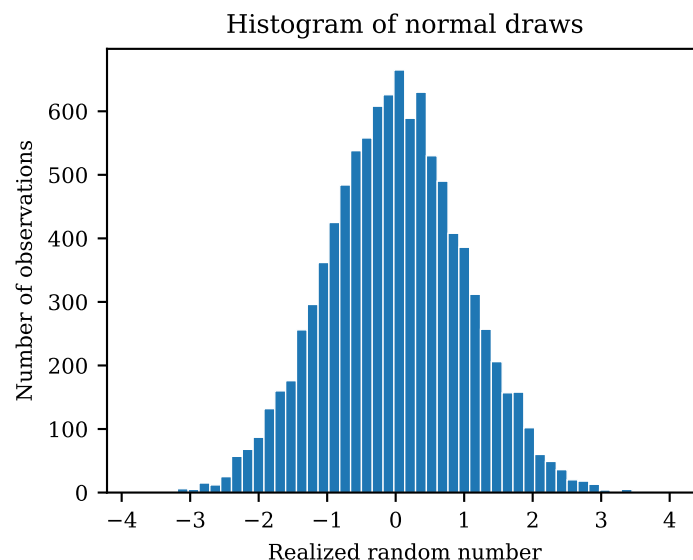
We use the `hist()` function to create histograms which can be used to nonparametrically visualize the distribution of some sample data.

To illustrate, we draw a random sample from the standard-normal distribution and visualize it using a histogram:

```
[9]: import matplotlib.pyplot as plt
import numpy as np

# Draw 10,000 standard-normal numbers
x = np.random.default_rng(seed=1234).normal(size=10_000)

# Plot the results as a histogram
plt.hist(x, bins=50, linewidth=0.5, edgecolor='white')
plt.xlabel('Realized random number')
plt.ylabel('Number of observations')
_ = plt.title('Histogram of normal draws')
```



Your turn

Load the house-price data from `../data/ames_houses.csv`. The column `SalePrice` contains sale prices in thousands of US dollars.

1. Create a histogram of `SalePrice` with 10 bins. Label the axes "Sale price (thousands of USD)" and "Number of houses".
2. Change the number of bins to 50 and rerun the plotting code, keeping the data unchanged.
3. Which plot reveals more detail? Why are the bars generally shorter with more bins?
4. Is the distribution approximately symmetric, or does it have a longer tail on one side?

1.4 Plotting categorical data

Instead of continuous numerical values on the x -axis, we can also plot categorical variables as bar charts using the function `bar()`.

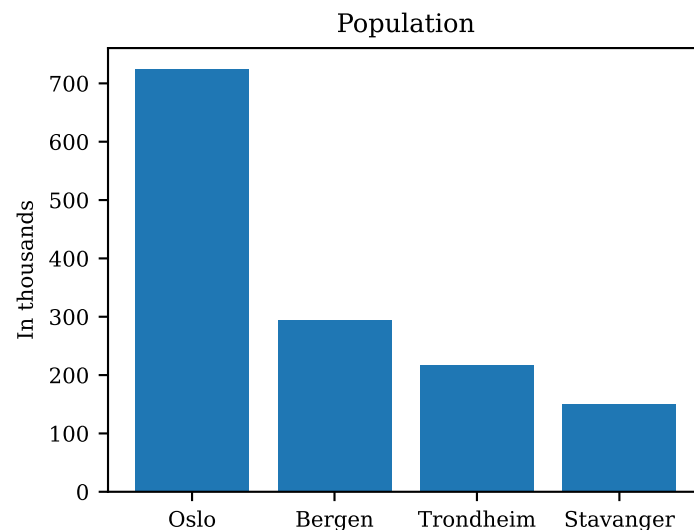
To illustrate, the following example creates a bar chart showing the population numbers of the four largest municipalities in Norway:

```
[10]: import matplotlib.pyplot as plt

# Define category labels
municipality = ['Oslo', 'Bergen', 'Trondheim', 'Stavanger']
# Population on January 1, 2025, in thousands
population = np.array([724290, 293709, 216518, 150123]) / 1000

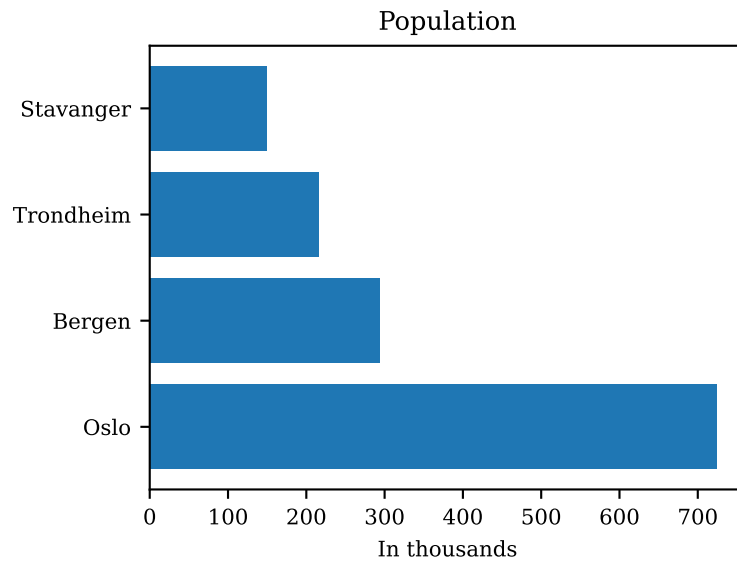
# Create bar chart
plt.bar(municipality, population)

# Add overall title
plt.title('Population')
_ = plt.ylabel('In thousands')
```



We use `barh()` to create *horizontal* bars:

```
[11]: plt.barh(municipality, population)
plt.title('Population')
_ = plt.xlabel('In thousands')
```

1.5 Adding labels and annotations

Matplotlib has numerous functions to add labels and annotations:

- Use `title()` and `suptitle()` to add titles. The latter adds a title for the whole figure, which might span multiple plots (axes).
- We can add axis labels by calling `xlabel()` and `ylabel()`.
- Set `label=` when plotting each dataset, then call `legend()` without arguments to build the legend from those labels.
- Use `text()` to add additional text at arbitrary locations.

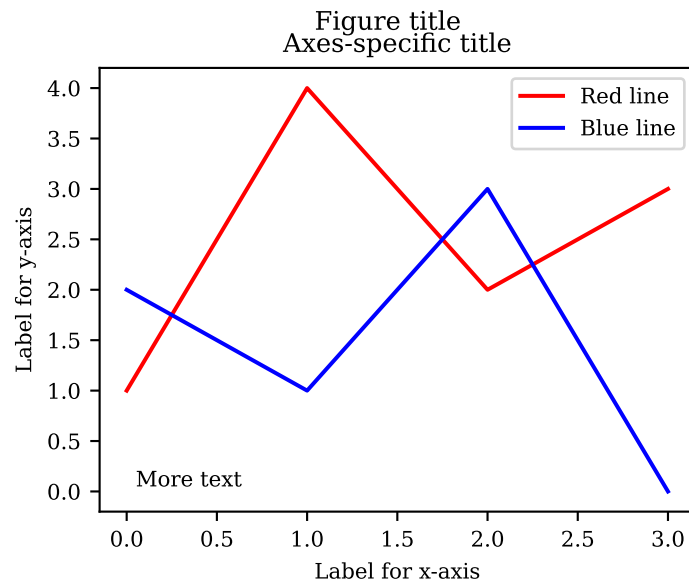
Example: Add title, labels, and a legend to some demo data

```
[12]: import matplotlib.pyplot as plt

# Demo data used for plotting
xvalues = [0, 1, 2, 3]
yvalues = [1, 4, 2, 3]
yvalues2 = [2.0, 1.0, 3.0, 0.0]

plt.plot(xvalues, yvalues, color='red')
plt.plot(xvalues, yvalues2, color='blue')

# Add titles, labels, legend, and text
plt.suptitle('Figure title')
plt.title('Axes-specific title')
plt.xlabel('Label for x-axis')
plt.ylabel('Label for y-axis')
# This labels-only form is discouraged: labels depend on plotting order.
# Prefer label= in plt.plot(), followed by plt.legend().
plt.legend(['Red line', 'Blue line'])
# Adds text at data coordinates (0.05, 0.05)
_ = plt.text(0.05, 0.05, 'More text')
```



Your turn

A bakery in Bergen sells skillingsboller (cinnamon buns) for NOK 25 each. Its fixed cost is NOK 60, and each bun costs NOK 10 to produce. For a quantity q produced and sold, revenue is $R(q) = 25q$ and total cost is $C(q) = 60 + 10q$.

1. Create a grid of 50 quantities from 0 to 10 using `np.linspace()`. Compute revenue and total cost, then plot both in the same graph.
2. Add a descriptive title and label the axes "Quantity" and "Revenue / cost (NOK)".
3. Set `label=` when plotting each line, then use `plt.legend()` to display a legend identifying "Revenue" and "Total cost".
4. The lines intersect at quantity 4 and revenue/cost NOK 100. Mark this point with a black dot.
5. Use `plt.text()` to place "Break-even" near this point without covering either line.

For which quantities does the bakery make a profit?

1.6 Plot limits, ticks and tick labels

We adjust the plot limits, ticks, and tick labels as follows:

- Plotting limits are set using the `xlim()` and `ylim()` functions. Each accepts a tuple (min, max) to set the desired range.
- Ticks and tick labels can be set by calling `xticks()` or `yticks()`.

Example: Setting plot limits, ticks, and tick labels

We demonstrate the use of these functions by plotting the sine function on the interval $[0, 2\pi]$:

```
[13]: import matplotlib.pyplot as plt
import numpy as np

# Create data using the sine function
xvalues = np.linspace(0.0, 2 * np.pi, 50)
```

```

yvalues = np.sin(xvalues)

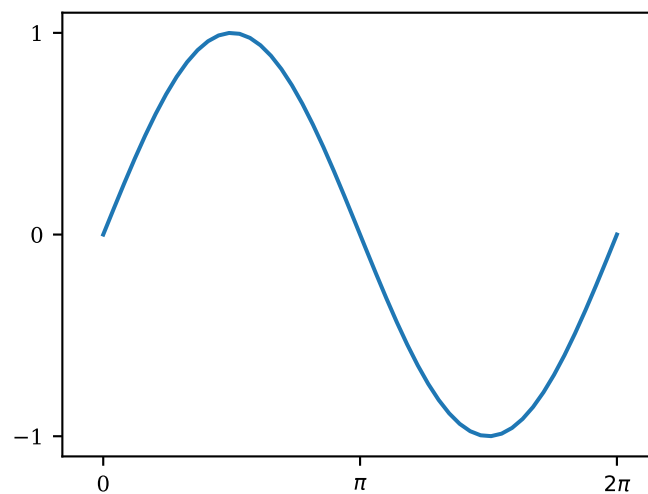
plt.plot(xvalues, yvalues)

# Set major ticks and labels for x-axis
# 1. Define tick positions
xticks = [0.0, np.pi, 2 * np.pi]
# 2. Define tick labels (can use LaTeX code!)
xtick_labels = ['0', r'$\pi$', r'$2\pi$']
plt.xticks(xticks, xtick_labels)

# Set major ticks for y-axis
plt.yticks([-1.0, 0.0, 1.0])

# Adjust plot limits in x and y direction
plt.xlim((-0.5, 2 * np.pi + 0.5))
_ = plt.ylim((-1.1, 1.1))

```



1.7 Adding straight lines

Quite often, we want to add horizontal or vertical lines to highlight a particular value. We can do this using the following functions:

- `axhline()` adds a *horizontal* line at a given *y*-value.
- `axvline()` adds a *vertical* line at a given *x*-value.
- `axline()` adds a line defined by two points or by a single point and a slope.

Example: Adding horizontal and vertical lines

Consider the sine function from above. We can add a horizontal line at 0 and two vertical lines at the points where the function attains its minimum and maximum as follows:

```

[14]: import matplotlib.pyplot as plt
import numpy as np

# Plot sine function (same as above)
xvalues = np.linspace(0.0, 2 * np.pi, 50)
yvalues = np.sin(xvalues)

plt.plot(xvalues, yvalues)

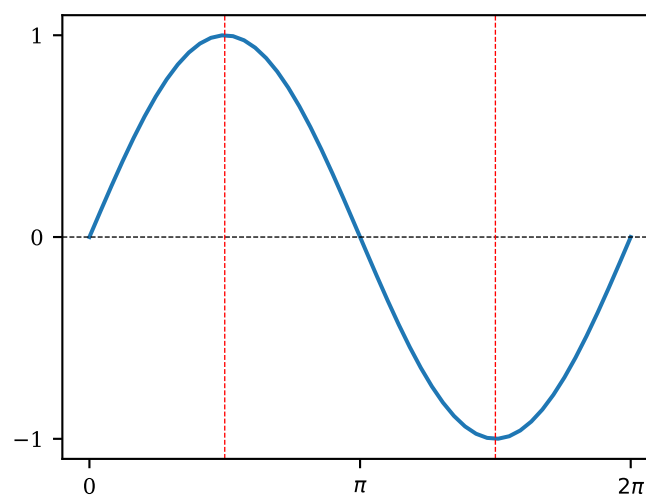
```

```
# Set major ticks and labels for x-axis
# 1. Define tick positions
xticks = [0.0, np.pi, 2 * np.pi]
# 2. Define tick labels (can use LaTeX code!)
xtick_labels = ['0', r'$\pi$', r'$2\pi$']
plt.xticks(xticks, xtick_labels)

# Set major ticks for y-axis
plt.yticks([-1.0, 0.0, 1.0])

# Add black dashed horizontal line at y-value 0
plt.axhline(0.0, lw=0.5, ls='--', c='black')

# Add red dashed vertical lines at maximum / minimum points
plt.axvline(0.5 * np.pi, lw=0.5, ls='--', c='red')
_ = plt.axvline(1.5 * np.pi, lw=0.5, ls='--', c='red')
```



Your turn

Plot the function $y = (x - 1)^2$ on the interval $[0, 2]$:

1. Create a grid of 50 x -values using `np.linspace()`.
2. Compute and plot the y -values.
3. Label the x - and y -axes with “ x ” and “ y ”.
4. Add the title “ $y = (x-1)^2$ ”.
5. Add a vertical line at $x = 1$ using a dashed line style.

1.8 Object-oriented interface

So far, we have only used the so-called pyplot interface which involves calling *global* plotting functions from `matplotlib.pyplot`. This interface is modeled on MATLAB and is convenient for quick plots.

We can instead use the object-oriented interface (called this way because we call methods of the [Figure](#) and [Axes](#) objects). Keeping explicit references to the figure and axes makes it easier to customize and reuse a plot, even when creating a single graph in a Jupyter notebook. The object-oriented interface is particularly useful when writing reusable code in Python files or creating a figure with multiple subplots.

To use the object-oriented interface, we need to get figure and axes objects. The easiest way to accomplish this is using the `subplots()` function, like this:

```
fig, ax = plt.subplots()
```

We then use methods of the Axes object returned by `subplots()` instead of the functions we have used so far. For example, instead of `plot()`, we use the `Axes.plot()` method.

See Matplotlib's [API guide](#) for a comparison of the pyplot and object-oriented interfaces.

As an example, we recreate the graph from the section on labels and annotations using the object-oriented interface:

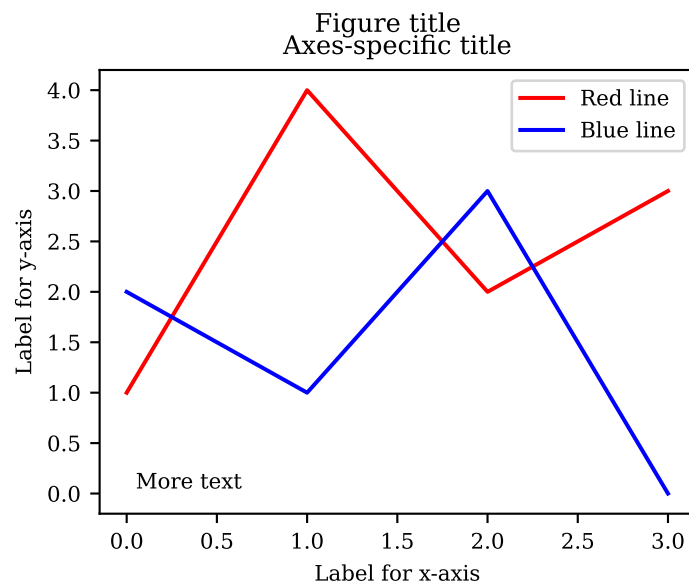
```
[15]: # Demo data used for plotting
xvalues = [0, 1, 2, 3]
yvalues = [1, 4, 2, 3]
yvalues2 = [2.0, 1.0, 3.0, 0.0]

[16]: import matplotlib.pyplot as plt

# Create Figure and Axes objects
fig, ax = plt.subplots()

ax.plot(xvalues, yvalues, color='red', label='Red line')
ax.plot(xvalues, yvalues2, color='blue', label='Blue line')

# Add titles, labels, legend, and text
fig.suptitle('Figure title') # Alternative: plt.suptitle()
ax.set_title('Axes-specific title') # Alternative: plt.title()
ax.set_xlabel('Label for x-axis') # Alternative: plt.xlabel()
ax.set_ylabel('Label for y-axis') # Alternative: plt.ylabel()
ax.legend() # Alternative: plt.legend()
# Adds text at data coordinates (0.05, 0.05)
_ = ax.text(0.05, 0.05, 'More text') # Alternative: plt.text()
```



The code is quite similar to the previous section, except that attributes are set using the `set_xxx()` methods of the `ax` object. For example, instead of calling `xlim()`, we use `ax.set_xlim()`.

Your turn

Recall the example from above in which we drew from a normal distribution and visualized the

sample using a histogram.

1. Repeat these steps, but create the histogram using the object-oriented Matplotlib interface.
2. Set the x-axis label to 'Realized random number' and the y-axis label to 'Number of observations'.
3. Add the title 'Histogram of normal draws'.

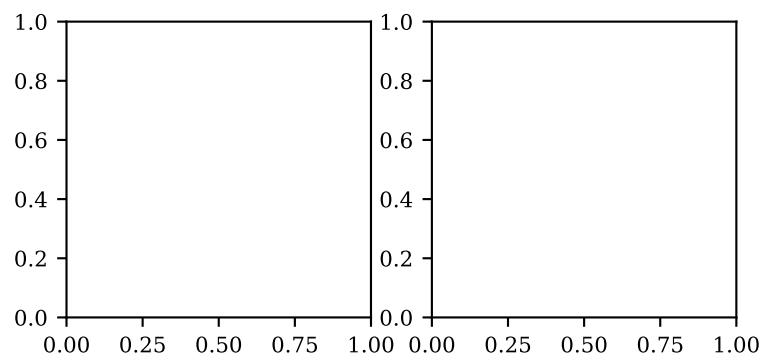
1.9 Working with multiple plots (axes)

The object-oriented interface becomes particularly useful if we want to create multiple axes (or figures). This can also be achieved using the pyplot programming model but is somewhat more obscure.

For example, to create a row with two subplots, we use:

```
[17]: import matplotlib.pyplot as plt

# Create one figure with 2 axes objects, arranged as two columns in a single row
fig, axes = plt.subplots(1, 2, figsize=(4.5, 2.0))
```



With multiple axes objects in a single figure (as in the above example), the axes object returned by `subplots()` is a NumPy array. Its elements map to the individual panels within the figure in a natural way.

We can visualize this mapping for the case of a single row and two columns as follows:

```
[18]: fig, axes = plt.subplots(1, 2, figsize=(3.5, 1.5))

for i, ax in enumerate(axes):
    # Turn off ticks of both axes
    ax.set_xticks(())
    ax.set_yticks(())
    # Label axes object
    text = f'axes[{i}]'
    ax.text(0.5, 0.5, text, va='center', ha='center', fontsize=18)
```

axes[0]

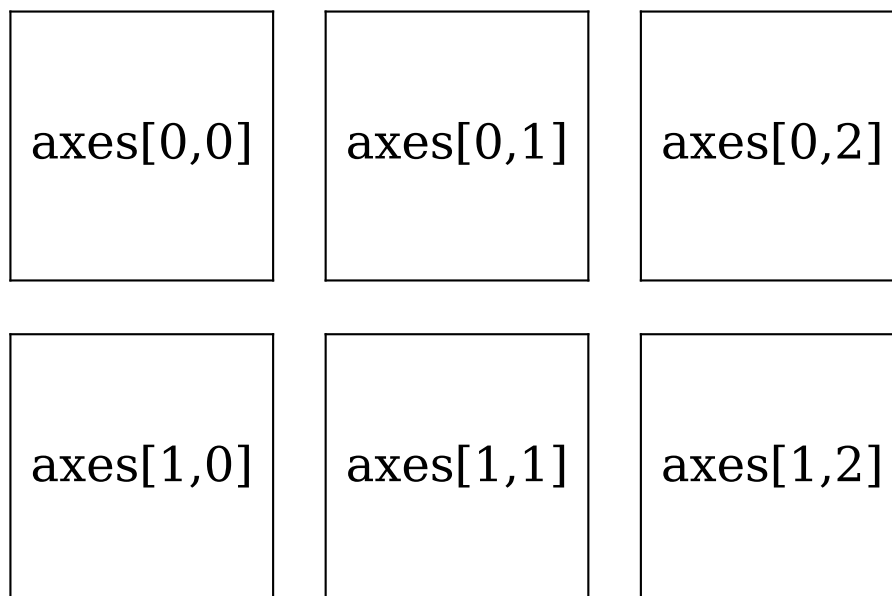
axes[1]

Don't worry about the details of how this graph is generated; the only takeaway here is how axes objects are mapped to the panels in the figure.

If we request panels in two dimensions, the axes object will be a 2-dimensional array, and the mapping of axes objects to panels will look like this instead:

```
[19]: # Create figure with 2 rows, 3 columns
nrow = 2
ncol = 3
fig, axes = plt.subplots(nrow, ncol, figsize=(6, 4))

for i in range(nrow):
    for j in range(ncol):
        # Get reference to current Axes object
        ax = axes[i, j]
        # Turn off ticks of both axes
        ax.set_xticks(())
        ax.set_yticks(())
        # Label axes object
        text = f'axes[{i},{j}]'
        ax.text(0.5, 0.5, text, va='center', ha='center', fontsize=18)
```



Example: Create a plot with 2 panels

We can use the elements of axes to plot into individual panels:

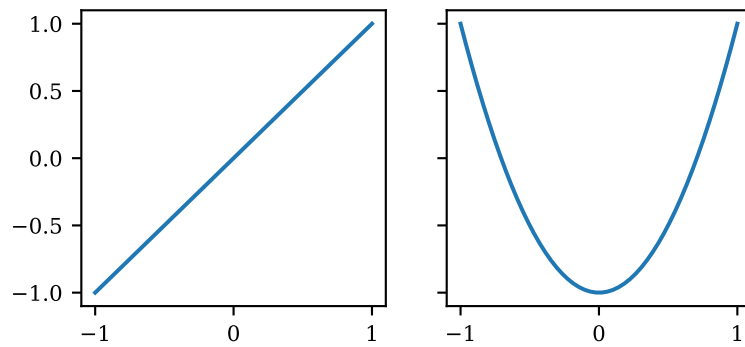
```
[20]: import matplotlib.pyplot as plt
import numpy as np

# Request a figure with 2 panels (1 row, 2 columns)
fig, axes = plt.subplots(1, 2, sharex=True, sharey=True, figsize=(4.5, 2.0))

xvalues = np.linspace(-1.0, 1.0, 50)

# Plot into first column
axes[0].plot(xvalues, xvalues)

# Plot into second column
_ = axes[1].plot(xvalues, 2 * xvalues**2.0 - 1)
```

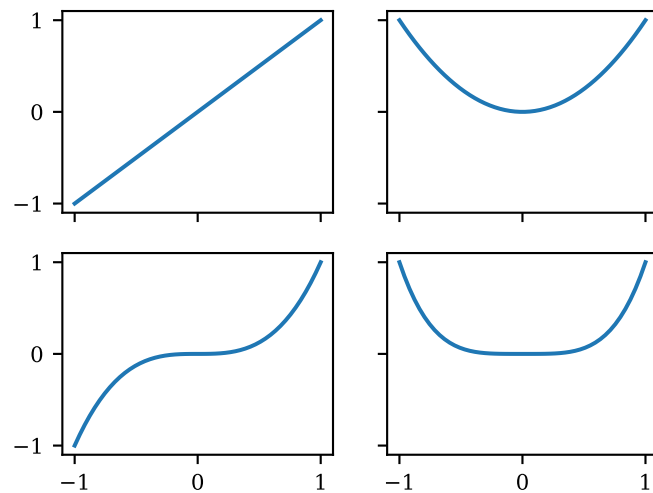


Example: Create a figure with 2 rows and 2 columns

```
[21]: # create figure with 2 rows, 2 columns
nrow = 2
ncol = 2
fig, axes = plt.subplots(nrow, ncol, sharex=True, sharey=True)

xvalues = np.linspace(-1.0, 1.0, 50)

# Plot the first four powers of x
exponent = 1
for i in range(nrow):
    for j in range(ncol):
        yvalues = xvalues**exponent
        axes[i, j].plot(xvalues, yvalues)
        # Increment exponent for next subplot
        exponent += 1
```



Note the use of `sharex=True` and `sharey=True`. This tells Matplotlib that all axes share the same plot limits, so the tick labels can be omitted in the figure's interior to preserve space.

Your turn

Create a figure with 3 columns (on a single row) and plot the following functions on the interval $[0, 6]$:

1. Subplot 1: $y = \sin(x)$
2. Subplot 2: $y = \sin(2 \cdot x)$

3. Subplot 3: $y = \sin(4 \cdot x)$

Hint: Use NumPy's `np.sin()` function.

1.10 Plotting with pandas

Pandas does not implement its own graphics library, but provides convenient wrappers around Matplotlib functions that can be used to quickly visualize data stored in DataFrames. Alternatively, we can extract the numerical data and pass it to Matplotlib's routines manually.

1.10.1 Bar charts

Let's return to our municipality population data. To plot population numbers as a bar chart, we can directly use pandas's `plot.bar()`:

```
[22]: import pandas as pd

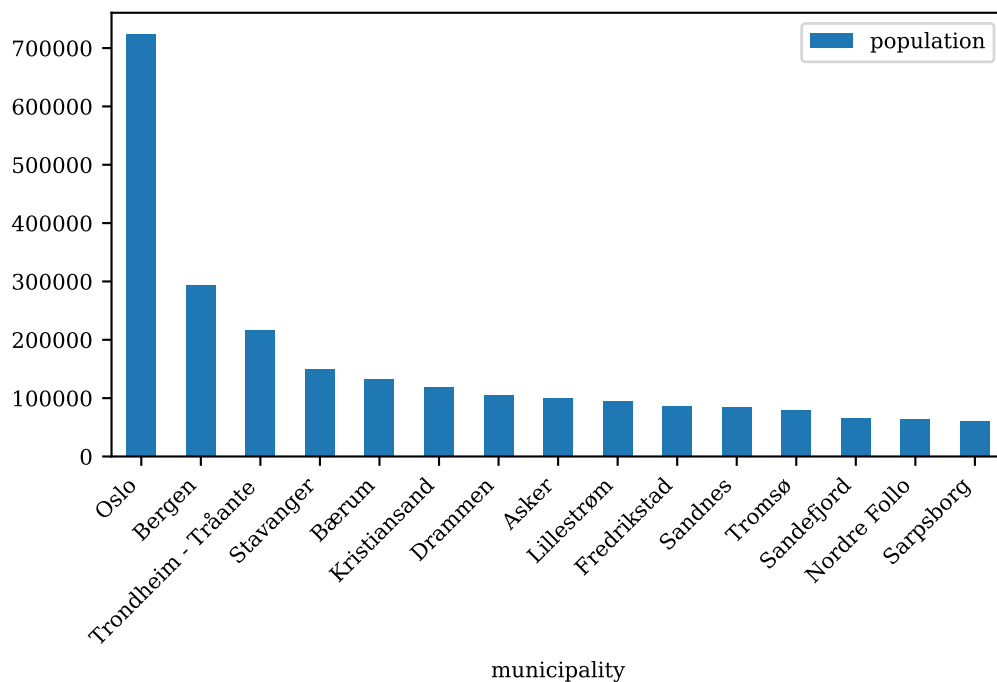
# Path to local data/ folder
DATA_PATH = '../data'

# Path to municipality data
filepath = f'{DATA_PATH}/ssb/municipalities.csv'

# Read only the municipality names and population numbers
df = pd.read_csv(filepath, usecols=['municipality', 'population'])

# Keep the 15 largest municipalities (the file is sorted by population)
df = df.iloc[:15]

# Create bar chart, specify figure size, rotate x-axis tick labels by 45 degrees
_ = df.plot.bar(x='municipality', y='population', rot=45, figsize=(6, 3))
_ = plt.xticks(ha='right')
```



Alternatively, we can construct the graph ourselves using Matplotlib:

```
[23]: import matplotlib.pyplot as plt

# Extract municipality names
labels = df['municipality']

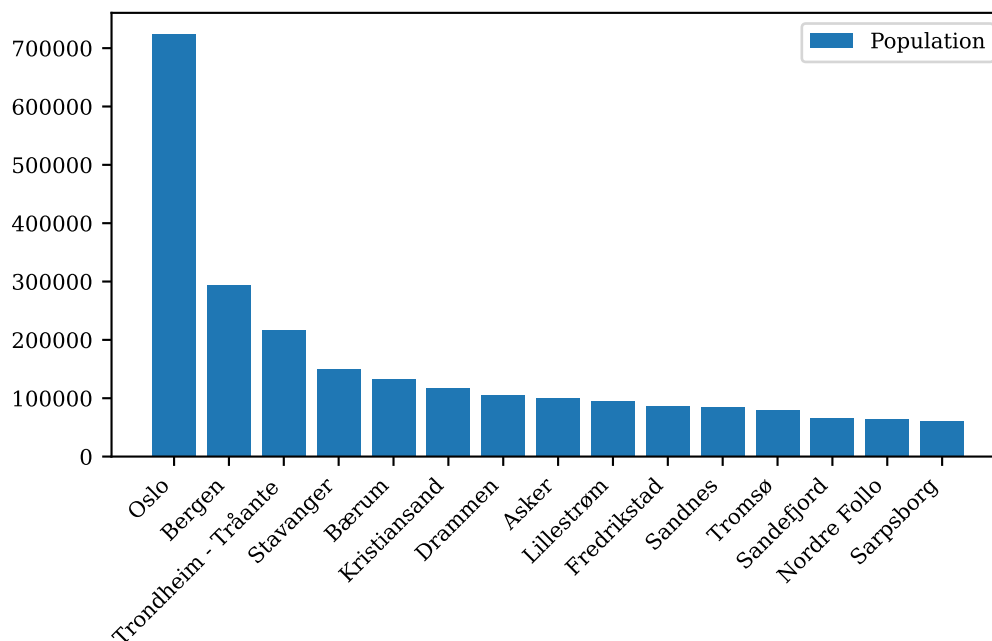
# Extract population numbers
values = df['population']

# Create new figure with desired size
plt.figure(figsize=(6, 3))

# Create bar chart
plt.bar(labels, values, label='Population')

# Add legend
plt.legend()

# Rotate tick labels by 45 degrees and align them with the bars
_ = plt.xticks(rotation=45, ha='right')
```



Matplotlib's functions usually work directly with pandas' data structures. In cases where they don't, we can convert a DataFrame or Series object to a NumPy array using the `to_numpy()` method.

1.10.2 Plotting time series data

To plot time series data, we can use the `DataFrame.plot()` method which optionally accepts arguments to specify which columns should be used for the x-axis and which for the y-axis. We illustrate this using the US unemployment rate at annual frequency.

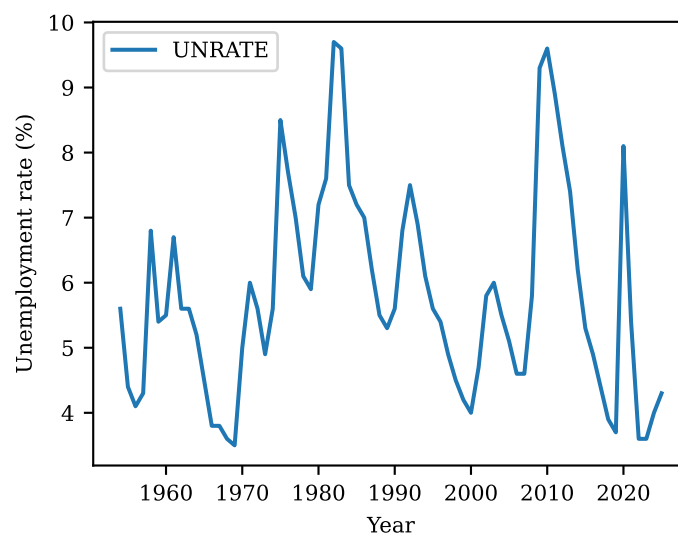
```
[24]: import pandas as pd

# Path to FRED data; DATA_PATH variable was defined above!
filepath = f'{DATA_PATH}/FRED/FRED_annual.csv'

# Read CSV data
```

```
df = pd.read_csv(filepath, sep=',')

# Plot unemployment rate by year
_ = df.plot(x='Year', y='UNRATE', ylabel='Unemployment rate (%)')
```



Your turn

Use the data files located in the folder `../..data/FRED` to perform the following tasks:

1. Load the macroeconomic time series data from `FRED_monthly_all.csv`. *Hint:* Use `pd.read_csv(..., parse_dates=['DATE'], index_col='DATE')` to automatically parse strings stored in the `DATE` column as dates and set `DATE` as the index.
2. Create a line plot, showing both the unemployment rate `UNRATE` and the inflation rate `INFLATION` in a single graph.

1.10.3 Scatter plots

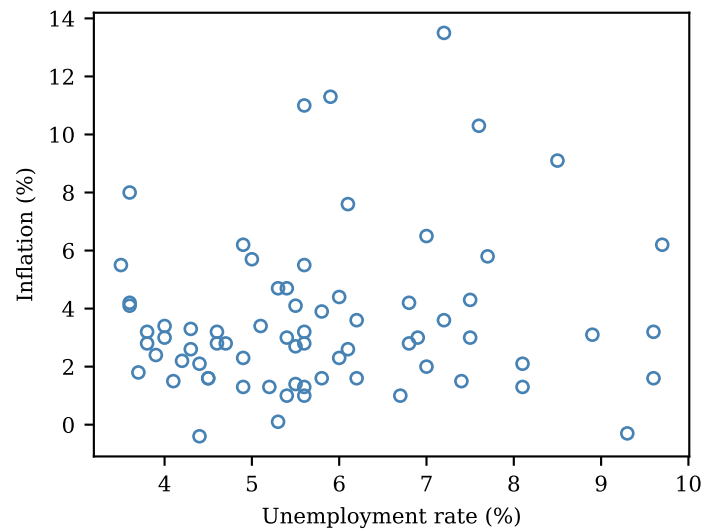
Using the `DataFrame.plot.scatter()` method, we can generate scatter plots, plotting one column against another. To illustrate, we plot US inflation against the unemployment rate in any given year over the post-war period.

Note that you can pass additional arguments (for example, `edgecolors`) to pandas' version of `scatter()`, which are passed on to Matplotlib's `scatter()`.

```
[25]: # Path to annual FRED data; DATA_PATH variable was defined above!
filepath = f'{DATA_PATH}/FRED/FRED_annual.csv'

# Read CSV data
df = pd.read_csv(filepath, sep=',')

_ = df.plot.scatter(
    x='UNRATE', # plot unemployment rate on x-axis
    y='INFLATION', # plot inflation rate on y-axis
    color='none',
    edgecolors='steelblue',
    xlabel='Unemployment rate (%)',
    ylabel='Inflation (%)',
)
```

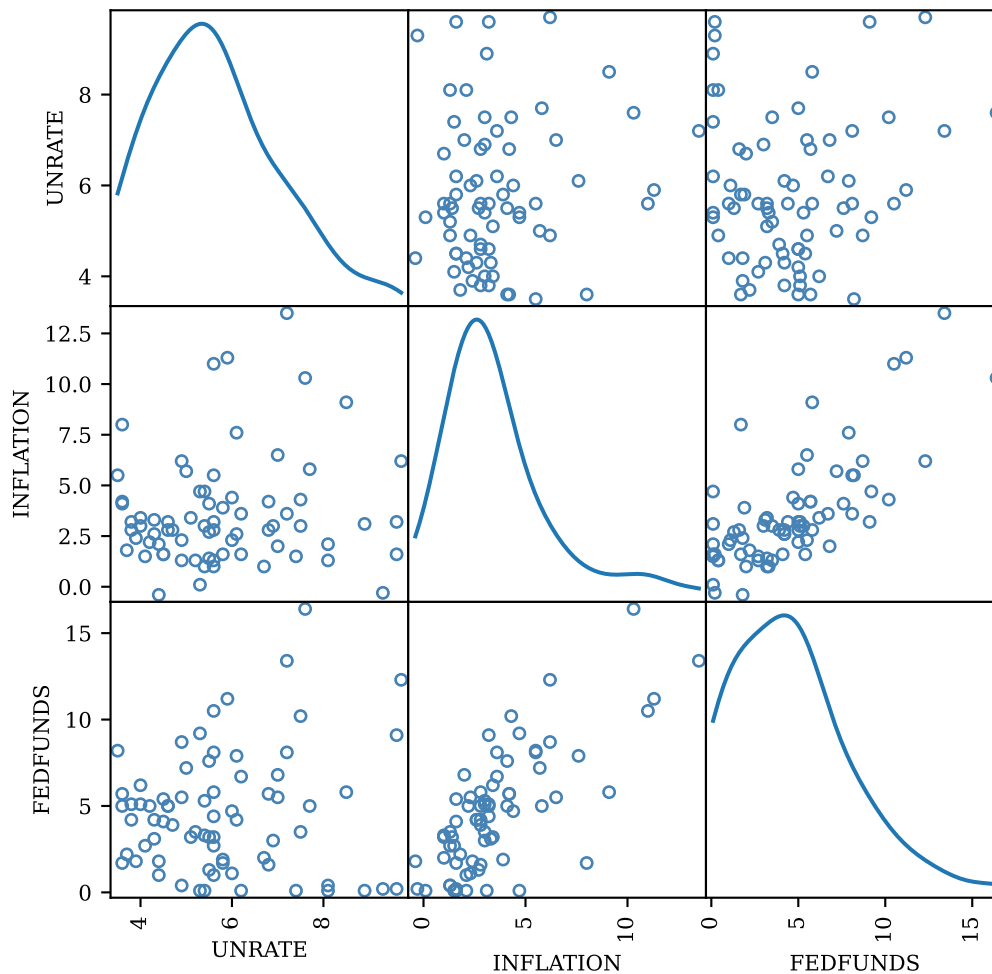


Pandas also offers the convenience function `scatter_matrix()` which lets us easily create pairwise scatter plots for more than two variables:

```
[26]: from pandas.plotting import scatter_matrix

# Columns to include in plot
columns = ['UNRATE', 'INFLATION', 'FEDFUNDS']

# Use argument diagonal='kde' to plot kernel density estimate
# in diagonal panels
ax = scatter_matrix(
    df[columns],
    figsize=(6, 6),
    diagonal='kde', # plot kernel density along diagonal
    s=70, # marker size
    color='none',
    edgecolors='steelblue',
    alpha=1.0,
)
```



1.10.4 Box plots

To quickly plot some descriptive statistics, we can use the `DataFrame.plot.box()` method provided by pandas. The box extends from the 25th percentile (Q_1) to the 75th percentile (Q_3), with a line at the median. The interquartile range is $IQR = Q_3 - Q_1$. By default, the whiskers extend to the most extreme observations within $1.5 \times IQR$ of the box. Points beyond the whiskers are plotted individually as potential outliers.

We demonstrate this by plotting the distribution of the unemployment rate, inflation, and the federal funds rate in the US:

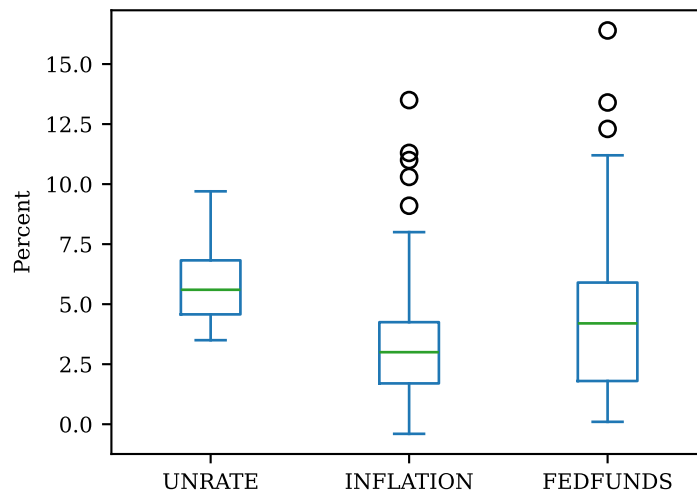
```
[27]: import pandas as pd

# Path to FRED data; DATA_PATH variable was defined above!
filepath = f'{DATA_PATH}/FRED/FRED_annual.csv'

# Read CSV data
df = pd.read_csv(filepath, sep=',')

# Include only the following columns in plot
columns = ['UNRATE', 'INFLATION', 'FEDFUNDS']

# Create box plot. Alternatively, use df.plot(kind='box')
_ = df[columns].plot.box(ylabel='Percent')
```



1.11 List of functions used in this lecture

The following list contains the central functions covered in this lecture. Each function name is linked to the official API documentation which you can consult for more details regarding function arguments and return values. The [go to section] links to the relevant section in this notebook.

1.11.1 Plotting functions

- `plot()` — plot data using lines [\[go to section\]](#)
- `scatter()` — create scatter plots [\[go to section\]](#)
- `hist()` — create histograms [\[go to section\]](#)
- `bar()` — create (vertical) bar charts [\[go to section\]](#)
- `barh()` — create *horizontal* bar charts [\[go to section\]](#)
- `axhline()` — adds a *horizontal* line at a given *y*-value [\[go to section\]](#)
- `axvline()` — adds a *vertical* line at a given *x*-value [\[go to section\]](#)

1.11.2 Plotting with pandas

- `DataFrame.plot()` — create a line plot (the default) from DataFrame contents [\[go to section\]](#)
- `DataFrame.plot.bar()` — create a bar chart from DataFrame contents [\[go to section\]](#)
- `DataFrame.plot.scatter()` — create a scatter plot from DataFrame contents [\[go to section\]](#)
- `scatter_matrix()` — create a *n*-by-*n* matrix of scatter plots from DataFrame contents [\[go to section\]](#)
- `DataFrame.plot.box()` — create a box plot from DataFrame contents [\[go to section\]](#)

1.11.3 Figures with multiple plots

- `subplots()` — create figure and (one or multiple) axes objects [\[go to section\]](#)

1.11.4 Annotations (labels, legends, ...)

- `title()` — add a title for a single subplot [\[go to section\]](#)
- `suptitle()` — add a title for the whole figure [\[go to section\]](#)
- `xlabel()` — add a label for the x -axis [\[go to section\]](#)
- `ylabel()` — add a label for the y -axis [\[go to section\]](#)
- `legend()` — add a legend [\[go to section\]](#)
- `text()` — add arbitrary text at some location [\[go to section\]](#)
- `xticks()` — specify ticks and tick labels for the x -axis [\[go to section\]](#)
- `yticks()` — specify ticks and tick labels for the y -axis [\[go to section\]](#)

1.11.5 Other customizations

- `xlim()` — specify plotting limits (lower and upper bounds) for the x -axis [\[go to section\]](#)
- `ylim()` — specify plotting limits (lower and upper bounds) for the y -axis [\[go to section\]](#)